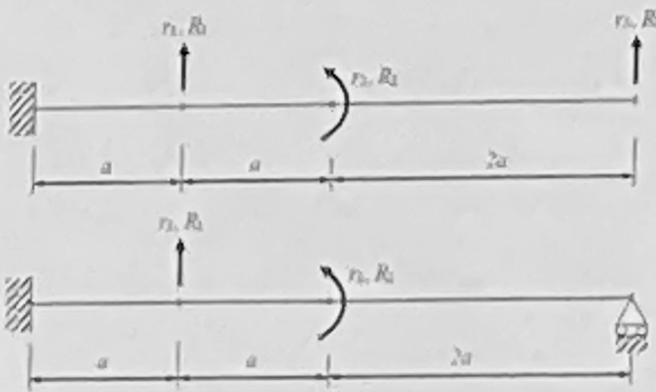
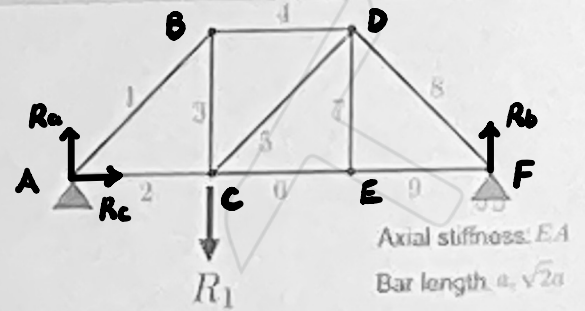


Problem to solve:

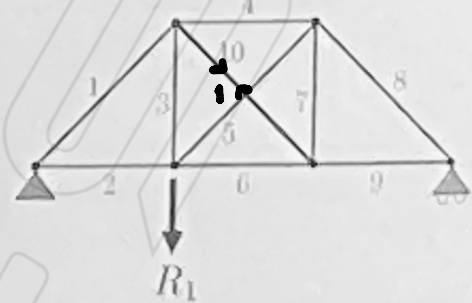


Given:

$$F = \frac{a}{6EI} \begin{bmatrix} 2a^2 & 3a & 11a^2 \\ 3a & 12 & 36a \\ 11a^2 & 36a & 128a^2 \end{bmatrix} \quad \text{Find the new } 2 \times 2 \text{ flexibility matrix for the structure.}$$



Determine the axial force acting on each bar.



Determine the force in the extra bar (bar 10)

1. For the statically indeterminate structure:

$$\begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} + \begin{bmatrix} F_{13} \\ F_{23} \end{bmatrix} R_3 = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\text{and } \begin{bmatrix} F_{31} & F_{32} \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} + F_{33} R_3 = 0$$

$$\therefore R_3 = - (F_{33})^{-1} \begin{bmatrix} F_{31} & F_{32} \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} - \begin{bmatrix} F_{13} \\ F_{23} \end{bmatrix} (F_{33})^{-1} \begin{bmatrix} F_{31} & F_{32} \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} = \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} - \begin{bmatrix} F_{13} \\ F_{23} \end{bmatrix} (F_{33})^{-1} \begin{bmatrix} F_{31} & F_{32} \end{bmatrix}$$

$$= \frac{a}{6EI} \left(\begin{bmatrix} 2a^2 & 3a \\ 3a & 12 \end{bmatrix} - \begin{bmatrix} 11a^2 \\ 36a \end{bmatrix} \frac{1}{128a^2} \begin{bmatrix} 11a^2 & 36a \end{bmatrix} \right)$$

$$= \frac{a}{6EI} \left(\begin{bmatrix} 2a^2 & 3a \\ 3a & 12 \end{bmatrix} - \begin{bmatrix} \frac{121}{128}a^2 & \frac{99}{32}a \\ \frac{99}{32}a & \frac{81}{8} \end{bmatrix} \right)$$

$$= \frac{a}{6EI} \begin{bmatrix} \frac{135}{128}a^2 & -\frac{3}{32}a \\ -\frac{3}{32}a & \frac{15}{8} \end{bmatrix}$$

$$= \frac{a}{768EI} \begin{bmatrix} 135a^2 & -12a \\ -12a & 240 \end{bmatrix}$$

$$2. \quad \sum F_h = R_c = 0$$

$$\sum F_v = R_a + R_b = R_1$$

$$\sum M(A) = R_1 a - 3R_b a = 0 \quad \therefore R_b = \frac{1}{3}R_1, \quad R_a = \frac{2}{3}R_1$$

$$\therefore \text{At } A, \quad \sum F_v = \frac{\sqrt{2}}{2}N_1 + R_a = 0 \quad \therefore N_1 = -\frac{2}{\sqrt{2}}R_1$$

$$\sum F_h = \frac{\sqrt{2}}{2}N_1 + N_2 = 0 \quad \therefore N_2 = \frac{2}{\sqrt{2}}R_1$$

$$\text{At B, } \Sigma F_v = N_3 + \frac{\sqrt{2}}{2} N_1 = 0 \quad \therefore N_3 = -\frac{2}{3} R_1$$

$$\Sigma F_h = N_4 - \frac{\sqrt{2}}{2} N_1 = 0 \quad \therefore N_4 = -\frac{2}{3} R_1$$

$$\text{At C, } \Sigma F_v = N_3 + \frac{\sqrt{2}}{2} N_5 - R_1 = 0 \quad \therefore N_5 = \frac{\sqrt{2}}{3} R_1$$

$$\Sigma F_h = N_2 - \frac{\sqrt{2}}{2} N_5 - N_6 = 0 \quad \therefore N_6 = \frac{1}{3} R_1$$

$$\text{At E, } \Sigma F_v = N_7 = 0 \quad \therefore N_7 = 0$$

$$\text{At F, } \Sigma F_v = \frac{\sqrt{2}}{2} N_8 + R_b = 0 \quad \therefore N_8 = -\frac{\sqrt{2}}{3} R_1$$

$$\Sigma F_h = N_9 + \frac{\sqrt{2}}{2} N_8 = 0 \quad \therefore N_9 = \frac{1}{3} R_1$$

When add bar 10, $\bar{N}_{10} = 1$

$$\Sigma F_h = R_c = 0$$

$$\Sigma F_v = R_u + R_b = 0$$

$$\Sigma M(A) = -3R_b = a \quad \therefore R_b = 0 \quad \therefore R_u = 0$$

$$\therefore \text{At A, } N_1 = 0, N_2 = 0$$

$$\text{At B, } \Sigma F_v = \frac{\sqrt{2}}{2} N_1 + N_3 + \frac{\sqrt{2}}{2} N_{10} = 0 \quad \therefore N_3 = -\frac{\sqrt{2}}{2}$$

$$\Sigma F_h = N_4 + \frac{\sqrt{2}}{2} N_{10} - \frac{\sqrt{2}}{2} N_1 = 0 \quad \therefore N_4 = -\frac{\sqrt{2}}{2}$$

$$\text{At C, } \Sigma F_v = N_3 + \frac{\sqrt{2}}{2} N_5 = 0 \quad \therefore N_5 = 1$$

$$\sum F_h = \frac{\sqrt{2}}{2} N_5 + N_6 - N_2 = 0 \quad \therefore N_6 = -\frac{\sqrt{2}}{2}$$

$$\text{At E, } \sum F_v = N_7 + \frac{\sqrt{2}}{2} N_{10} = 0 \quad \therefore N_7 = -\frac{\sqrt{2}}{2}$$

$$\text{At F, } N_8 = 0, N_9 = 0$$

$$\therefore \delta_{10} = \sum \frac{N_b^{10} \bar{N}_b^{10}}{EA} L_b$$

$$= (0 + 0 - \frac{1}{3}a + \frac{1}{3}a + \frac{2}{3}a - \frac{\sqrt{2}}{6}a + 0 + 0 + 0 + 0) R,$$

$$= (\frac{4 - \sqrt{2}}{6} a) R,$$

$$\bar{\delta}_{10} = \sum \frac{N_b^{10} \bar{N}_b^{10}}{EA} L_b$$

$$= 0 + 0 + \frac{1}{2}a + \frac{1}{2}a + \sqrt{2}a + \frac{1}{2}a + \frac{1}{2}a + 0 + 0 + \sqrt{2}a$$

$$= (2 + 2\sqrt{2}) a$$

$$\therefore \delta_{10} + X \bar{\delta}_{10} = 0$$

$$\therefore N_{10} = X = -\frac{\delta_{10}}{\bar{\delta}_{10}} = \frac{-6 + 5\sqrt{2}}{12} R, \quad \frac{6 - 5\sqrt{2}}{12} R,$$